ComSIA-2027
International Conference on Computing Systems and Intelligent Applications

Organized **by School of Open Learning, University of Delhi & Shaheed Rajguru College of Applied Sciences for Women, University of Delhi**
5-6th March 2027.

********** CALL FOR PAPERS **********

SPECIAL SESSION ON

Cybersecurity and Resilience in Next-Generation Networks and Computing Systems

SESSION ORGANIZERS:

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EDITORIAL BOARD: (Optional)

To be confirmed

SESSION DESCRIPTION:

6G and beyond-5G networks are evolving into AI-native, cloud-edge integrated, and highly programmable infrastructures. These networks will support autonomous mobility, smart cities, Industry 5.0, digital healthcare, massive Internet of Things deployments, and critical services through technologies such as network slicing, software-defined networking, non-terrestrial networks, semantic communications, digital twins, and integrated sensing and communication. Their open, distributed, and heterogeneous architecture, however, expands the attack surface and creates new challenges in security, privacy, trust, resilience, interoperability, and sustainable operation.

This special session will bring together researchers, academicians, industry professionals, and practitioners to discuss advances in securing and strengthening AI-native next-generation networks. It welcomes theoretical contributions, novel architectures and algorithms, experimental studies, testbeds, datasets, standards-oriented work, and real-world case studies. Particular emphasis is placed on trustworthy artificial intelligence, zero-trust networking, privacy-preserving learning, post-quantum security, intelligent threat detection, and resilient operation across terrestrial and non-terrestrial 6G environments. Original research articles, review papers, and application-oriented studies addressing secure, dependable, and sustainable future networks are encouraged.

RECOMMENDED TOPICS:

Topics to be discussed in this special session include (but are not limited to) the following:

- AI-native architectures and autonomous security for B5G and 6G networks
- Trustworthy, explainable, and adversarially robust AI for network security
- Zero-trust architecture, continuous authentication, and adaptive access control
- Intrusion, anomaly, malware, botnet, and DDoS detection using AI
- Federated, split, distributed, and privacy-preserving learning
- Post-quantum cryptography and quantum-safe security for future networks
- Physical-layer security, lightweight cryptography, and secure waveform design
- Security and privacy for network slicing, SDN, NFV, and open RAN
- Secure edge, fog, cloud, and multi-access edge computing
- Blockchain, decentralized identity, and distributed trust management
- Security of IoT, IIoT, cyber-physical systems, and Industry 5.0
- Resilient vehicular, drone, satellite, and non-terrestrial networks
- Security for semantic communications and integrated sensing and communication
- Digital twins for network monitoring, threat intelligence, and cyber resilience
- Secure orchestration, resource allocation, and spectrum sharing
- Generative AI and large language models for network operations and cybersecurity
- Energy-efficient, green, and sustainable security mechanisms
- Threat modeling, attack simulation, cyber ranges, and digital forensics
- Security standards, governance, compliance, datasets, testbeds, and deployments

SUBMISSION PROCEDURE:

Researchers and practitioners are invited to submit original papers to the special session on "Cybersecurity and Resilience in Next-Generation Networks and Computing Systems." Submissions must not be under review or published elsewhere. Authors should follow the conference manuscript-submission guidelines available at https://comsia.in/paper_submission.html.

All submissions will undergo double-blind peer review in accordance with the conference review process.

NOTE: When submitting a paper to this special session, please specify "Cybersecurity and Resilience in Next-Generation Networks and Computing Systems" at the top of the first page, above the paper title.

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